

Introduction to Bioinformatics

Day 2: Sequence Alignment and
Phylogenetic
13th January 2026

Outline Day 2

Morning (9-12 pm):

Pairwise and Multiple Sequence Alignment:

- **Overview of alignment concepts:** Pairwise alignment (global vs local) and multiple sequence alignment
- **Algorithm**
 - **Global Alignments:** Needleman-Wunsch algorithm
 - **Local Alignments:** Smith-Waterman algorithm
 - **Multiple Sequence Alignment:** ClustalW and Clustal Omega

Practical Session:

Primary option:

- Use NCBI BLAST (web-based) to perform local alignment and sequence similarity searches
- Perform multiple sequence alignment using Clustal Omega (web-based)
- Code with python

Afternoon (2-5 pm): Phylogenetics

Phylogenetics:

- Introduction to Phylogenetics tree concept, and significance in evolution biology.

Practical Session:

- Construct phylogenetics trees using Clustal Omega (web-based) starting from multiple sequence alignments.

Code with python

Day 2: Sequence Alignment and Phylogenetic

Morning Session

What is a Sequence?

Specific linear order of nucleotides (in DNA or RNA) or amino acids (in proteins) that determines the structure and function of these biomolecules.

DNA Sequence: The arrangement of bases (adenine [A], thymine [T], cytosine [C], guanine [G]).

RNA Sequence: The arrangement of bases (adenine [A], uracil [U], cytosine [C], guanine [G]).

Protein Sequence: The arrangement of amino acids in a polypeptide chain.

Sequences encode genetic information and are fundamental to biological processes like replication, transcription, and translation.



Sample genetic code with complementary strands.

© G.Osuri

Examples of Sequences?

DNA Sequence: order of nucleotide bases (adenine, thymine, cytosine, guanine)

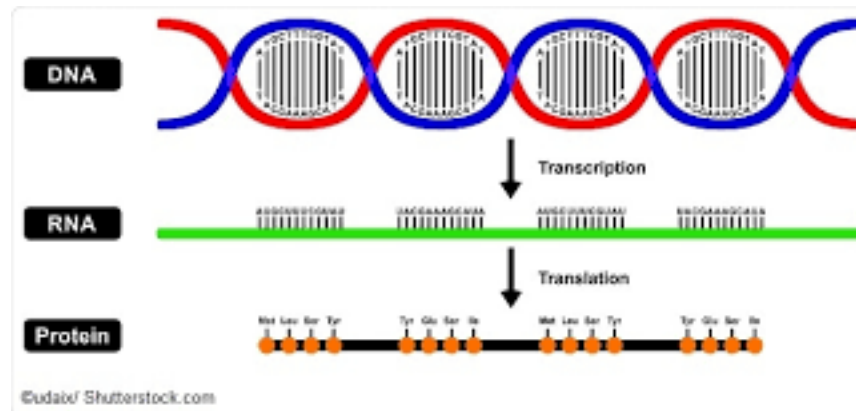
Example. **ATCGTACGGA**

RNA Sequence: Similar to, DNA1 but thymine (T) is replaced by uracil (U) 'in RNA.

Example: **AUCGAUCGGA**

Protein Sequence: order ,of amino acids (e.g., methionine, alanine, leucine,,glycine)

Example· MET-ALA-LEU-GLV



FASTA

A text-based file format used to represent nucleotide or protein sequences. It is commonly used in bioinformatics to store and exchange sequence data. A FASTA file consists of:

- 1. Header Line:** Starts with a > symbol, followed by an identifier or description of the sequence.
- 2. Sequence Data:** The following lines contain the actual sequence, either nucleotide (DNA/ RNA) or protein, and can span multiple lines.

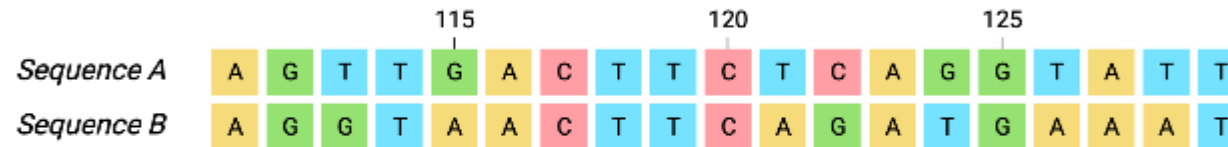
Example:

```
>Sequence_ID_1 Description of the first sequence  
AGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTA
```

```
>Sequence_ID_2 Description of the second sequence  
TGCATGCATGCATGCATGCGTACGATCGTACG
```

What is Sequence Alignment?

Sequence alignment is the process of arranging two or more biological sequences such as nucleotide (DNA, RNA), or amino acid (protein) sequences to identify regions of similarity.



What alignments can help?



DETERMINE EVOLUTIONARY
RELATIONSHIPS AMONG GENES,
PROTEINS, AND SPECIES



DETERMINE FUNCTION OF A
NEWLY DISCOVERED GENE
SEQUENCE

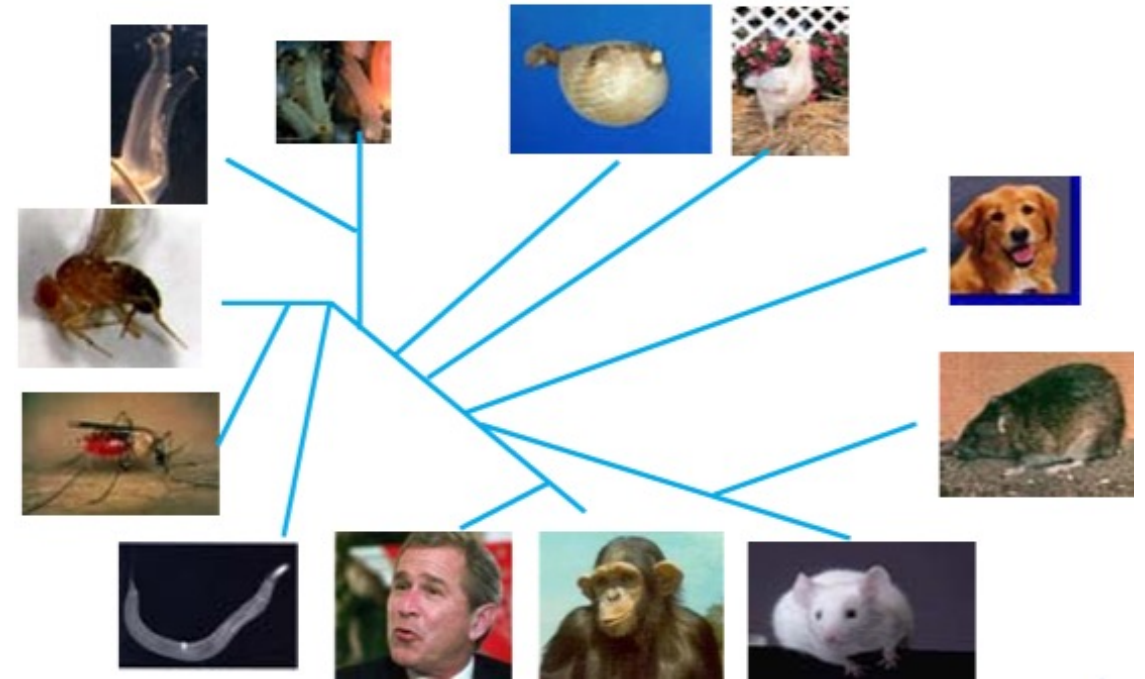


PREDICTING STRUCTURE AND
FUNCTION OF PROTEIN

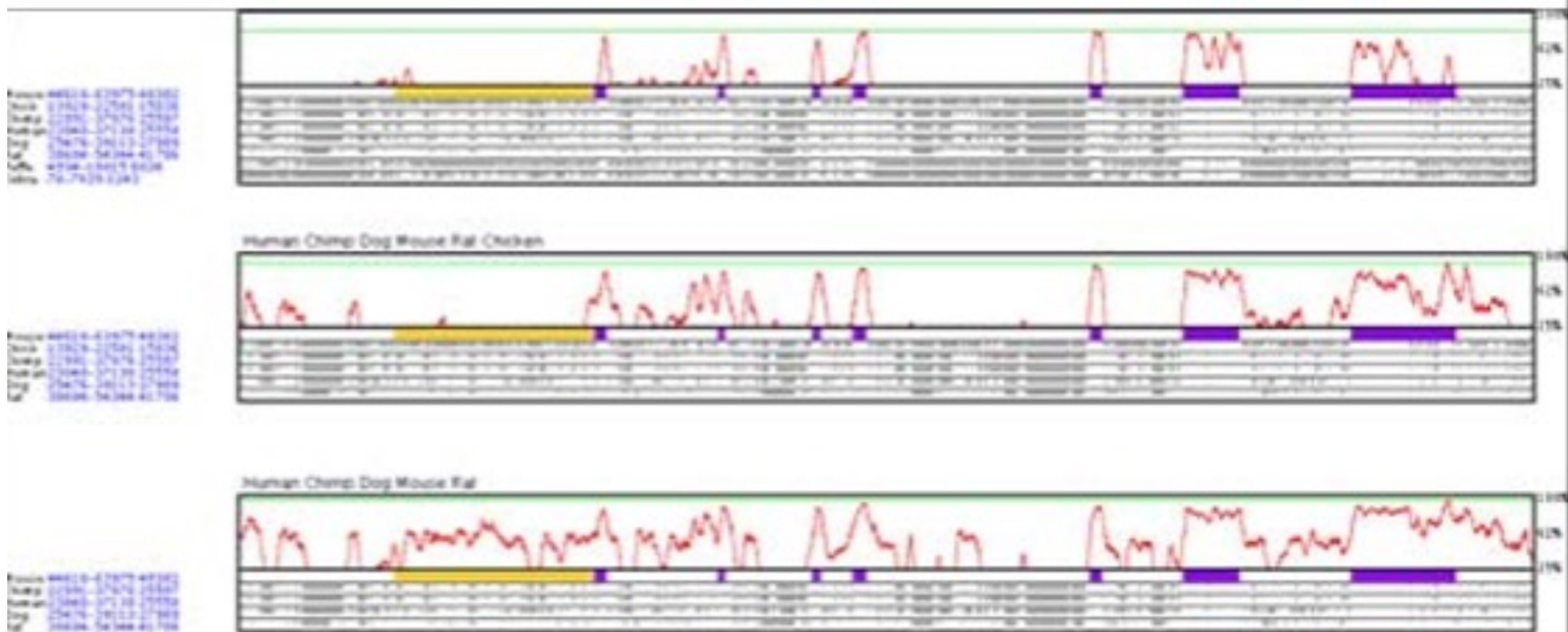
Importance of Sequence Alignment in Bioinformatics

Evolutionary Relationships

- Helps in identifying similarities between genes/
proteins from different species.
- Homologous sequences, (genes from a common ancestor) can be identified through alignment.
- Helps inferring evolutionary distances using substitution patterns (changes in nucleotides).
- Helps in phylogenetic tree construction



Sequence conversation implies function



Alignment is the key to

- Finding important regions
- Determining function
- Uncovering the evolutionary forces



Algorithm used in Dynamic programming

Needleman-wunsch Algorithm

**Developed by Saul.b. needleman
& christian.d. wunsch**

Referred as global alignment

**Used in aligning 2 closely related
sequence**

Compares the whole sequence

**Tools:- EMBOSS-Needle,
Specialised BLAST**

Smith-Waterman algorithm

**Developed by Temple.F.Smith
& Michael. S. Waterman**

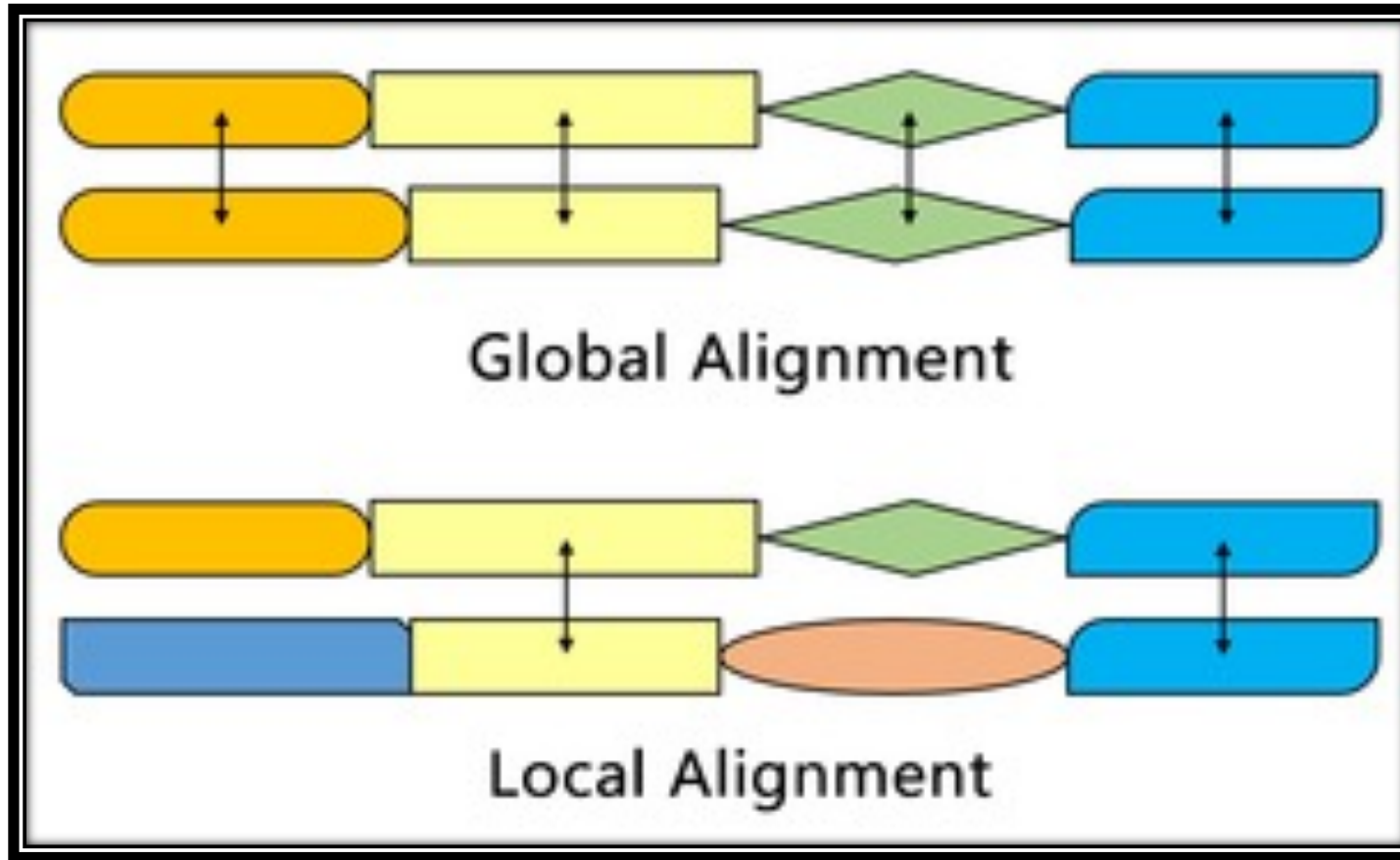
Referred as local alignment

**Used in aligning divergent
sequences**

**Compares a patch from the
sequence**

**Tools:-EMBOSS-Water,
LALIGN**

Comparison of Global and Local alignment in general



Needleman-Wunsch algorithm

- Aligns protein or nucleic acid sequences
- Divides a large problem into a series of smaller problems
- Uses the solution of smaller problem to reconstruct a solution to the larger problems

Constructing the matrix

We will have 2 matrices of 2D representation viz,

1. The score matrix
2. Traceback matrix

The N-W algorithm consists of 4 steps:-

1. Initialization of the score matrix
2. Filling up the matrix
3. Traceback
4. Alignment

1. Initializing the scoring matrix

	-	G	T	A	C	G	C
-	0	-2	-4	-6	-8	-10	-12
G	-2						
C	-4						
T	-6						
A	-8						
G	-10						
C	-12						

Parameter	Value
Sequence 1	G C T A G C
Sequence 2	G T A C G C
Match Score	+1
Mismatch Score	-1
Gap Penalty	-2

2. Filling the matrix

		G
	0	-2
G	-2	X

For x :

This cell has 3 possible values

- Top :- $(-2)+(-2) = -4$
- Left :- $(-2)+(-2) = -4$
- Top-left :- $(0)+(1) = 1$

The highest value is 1 and thus it is entered into the cell
i.e $x = 1$

		G
	0	-2
G	-2	1

Contd..

		G	C
	0	-2	-4
G	-2	1	X
C	-4	Y	

For **X** :

Top: $(-4)+(-2) = (-6)$

Left: $(+1)+(-2) = (-1)$

Top-Left: $(-2)+(-1) = (-3)$

For **Y** :

Top: $(1)+(-2) = (-1)$

Left: $(-4)+(-2) = (-6)$

Top-Left: $(-2)+(-1) = (-3)$

The highest value for X and Y is
-1, thus it is entered into the cell.
i.e **X = -1; Y = -1**

		G	C
	0	-2	-4
G	-2	1	-1
C	-4	-1	

1. Initializing the scoring matrix

	-	G	T	A	C	G	C
-	0	-2	-4	-6	-8	-10	-12
G	-2	1	-1	-3	-5	-7	-9
C	-4						
T	-6						
A	-8						
G	-10						
C	-12						

Parameter	Value
Sequence 1	G C T A G C
Sequence 2	G T A C G C
Match Score	+1
Mismatch Score	-1
Gap Penalty	-2

1. Initializing the scoring matrix

	-	G	T	A	C	G	C
-	0	-2	-4	-6	-8	-10	-12
G	-2	1	-1	-3	-5	-7	-9
C	-4	-1	0	-2	-2	-4	-6
T	-6						
A	-8						
G	-10						
C	-12						

Parameter	Value
Sequence 1	G C T A G C
Sequence 2	G T A C G C
Match Score	+1
Mismatch Score	-1
Gap Penalty	-2

1. Initializing the scoring matrix

	-	G	T	A	C	G	C
-	0	-2	-4	-6	-8	-10	-12
G	-2	1	-1	-3	-5	-7	-9
C	-4	-1	0	-2	-2	-4	-6
T	-6	-3	0	-1	-3	-3	-5
A	-8						
G	-10						
C	-12						

Parameter	Value
Sequence 1	G C T A G C
Sequence 2	G T A C G C
Match Score	+1
Mismatch Score	-1
Gap Penalty	-2

1. Initializing the scoring matrix

	-	G	T	A	C	G	C
-	0	-2	-4	-6	-8	-10	-12
G	-2	1	-1	-3	-5	-7	-9
C	-4	-1	0	-2	-2	-4	-6
T	-6	-3	0	-1	-3	-3	-5
A	-8	-5	-2	1	-1	-3	-4
G	-10						
C	-12						

Parameter	Value
Sequence 1	G C T A G C
Sequence 2	G T A C G C
Match Score	+1
Mismatch Score	-1
Gap Penalty	-2

1. Initializing the scoring matrix

	-	G	T	A	C	G	C
-	0	-2	-4	-6	-8	-10	-12
G	-2	1	-1	-3	-5	-7	-9
C	-4	-1	0	-2	-2	-4	-6
T	-6	-3	0	-1	-3	-3	-5
A	-8	-5	-2	1	-1	-3	-4
G	-10	-7	-4	-1	0	0	-2
C	-12						

Parameter	Value
Sequence 1	G C T A G C
Sequence 2	G T A C G C
Match Score	+1
Mismatch Score	-1
Gap Penalty	-2

1. Initializing the scoring matrix

	-	G	T	A	C	G	C
-	0	-2	-4	-6	-8	-10	-12
G	-2	1	-1	-3	-5	-7	-9
C	-4	-1	0	-2	-2	-4	-6
T	-6	-3	0	-1	-3	-3	-5
A	-8	-5	-2	1	-1	-3	-4
G	-10	-7	-4	-1	0	0	-2
C	-12	-9	-6	-3	0	-1	1

Parameter	Value
Sequence 1	G C T A G C
Sequence 2	G T A C G C
Match Score	+1
Mismatch Score	-1
Gap Penalty	-2

3. Traceback

	-	G	T	A	C	G	C
-	0	-2	-4	-6	-8	-10	-12
G	-2	1	-1	-3	-5	-7	-9
C	-4	-1	0	-2	-2	-4	-6
T	-6	-3	0	-1	-3	-3	-5
A	-8	-5	-2	1	-1	-3	-4
G	-10	-7	-4	-1	0	0	-2
C	-12	-9	-6	-3	0	-1	1

Step	Position	Seq1	Seq2	Action	Score Contribution
1	(6,6)	C	C	Match	+1
2	(5,5)	G	G	Match	+1
3	(4,4)	A	C	Mismatch	-1
4	(4,3)	—	A	Gap	-2
5	(3,3)	T	T	Match	+1
6	(2,2)	C	—	Gap	-2
7	(1,1)	G	G	Match	+1

4. Alignment

Rules :-

1. If arrow is vertical/horizontal assign a gap and a character

↑ / → :- Gaps and characters
Where to Assign
a gap and a character ?

Sequenc
e 1 or 2



Contd...

Ans) The gap will be assigned in the direction of the arrow and the character will be assigned in the opposite direction

2. If there is a diagonal arrow both the characters will be assigned

:- Both characters



Result of alignment

	-	G	T	A	C	G	C
-	0	-2	-4	-6	-8	-10	-12
G	-2	1	-1	-3	-5	-7	-9
C	-4	-1	0	-2	-2	-4	-6
T	-6	-3	0	-1	-3	-3	-5
A	-8	-5	-2	1	-1	-3	-4
G	-10	-7	-4	-1	0	0	-2
C	-12	-9	-6	-3	0	-1	1

Metric	Value
Optimal Score	1
Number of Matches	4
Number of Mismatches	1
Number of Gaps	1
Aligned Seq1	G C T A G C
Aligned Seq2	G - T C G C

Smith-Waterman algorithm

- S-W algorithm is modified version of Needleman-Wunsch algorithm
- Negative scoring matrix cells are set to zero
- Traceback procedure starts at the highest scoring matrix cells and procedure until a cell with score zero is found

Constructing the matrix

	–	G	T	A	C	G	C
–	0	0	0	0	0	0	0
G	0	.	.	.	.	.	.
C	0	.	.	.	.	.	.
T	0	.	.	.	.	.	.
A	0	.	.	.	.	.	.
G	0	.	.	.	.	.	.
C	0	.	.	.	.	.	.

Parameter	Value
Sequence 1	G C T A G C
Sequence 2	G T A C G C
Match Score	+1
Mismatch Score	–1
Gap Penalty	–2

Contd..

		G	C
	0	0	0
G	0	1	X
C	0	Y	

For X:
Top: $(0) + (-2) = -2$
Left: $(1) + (-2) = -1$
Top-Left: $(0) + (-1) = -1$
Zero: 0
For Y:
Top: $(1) + (-2) = -1$
Left: $(0) + (-2) = -2$
Top-Left: $(0) + (-1) = -1$
Zero: 0

		G	C
	0	0	0
G	0	1	0
C	0	0	

Constructing the matrix

	–	G	T	A	C	G	C
–	0	0	0	0	0	0	0
G	0	1	0	0	0	1	0
C	0	.	.	.	.	.	.
T	0	.	.	.	.	.	.
A	0	.	.	.	.	.	.
G	0	.	.	.	.	.	.
C	0	.	.	.	.	.	.

Parameter	Value
Sequence 1	G C T A G C
Sequence 2	G T A C G C
Match Score	+1
Mismatch Score	–1
Gap Penalty	–2

Constructing the matrix

	-	G	T	A	C	G	C
-	0	0	0	0	0	0	0
G	0	1	0	0	0	1	0
C	0	0	0	0	1	0	2
T	0	0	1	0	0	0	0
A	0	0	0	2	0	0	0
G	0	1	0	0	1	1	0
C	0	0	0	0	1	0	2

Parameter	Value
Sequence 1	G C T A G C
Sequence 2	G T A C G C
Match Score	+1
Mismatch Score	-1
Gap Penalty	-2

Global maximum encountered: 2 (multiple cells: (2,6), (4,3), (6,6)).

Traceback

	-	G	T	A	C	G	C
-	0	0	0	0	0	0	0
G	0	1	0	0	0	1	0
C	0	0	0	0	1	0	2
T	0	0	1	0	0	0	0
A	0	0	0	2	0	0	0
G	0	1	0	0	1	1	0
C	0	0	0	0	1	0	2

Alignment

Seq1: G C T A [G C]

Seq2: G T A C [G C]

Alignment

Seq1: [G C] T A G C

Seq2: G T A C [G C]

Alignment

Seq1: G C [T A] G C

Seq2: G [T A] C G C

Alignment

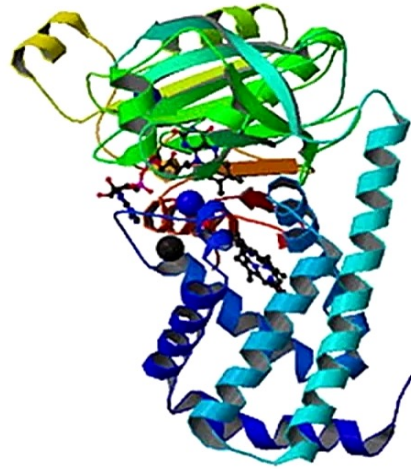
Rules:

Same as of Needleman- Wunsch algorithm

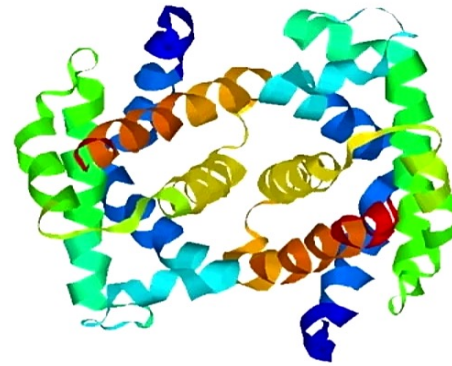
Difference between the procedure of S-W and N-W algorithm

	Smith– Waterman algorithm	Needleman– Wunsch algorithm
Initialization	First row and first column are set to 0	First row and first column are subject to gap penalty
Scoring	Negative score is set to 0	Score can be negative
Traceback	Begin with the highest score, end when 0 is encountered	Begin with the cell at the lower right of the matrix, end at top left cell

Example



Flavohemoprotein of
Escherichia coli



haemoglobin subunit alpha
Homo sapiens



MNPYIYLGGAILAEVIGTTLMKFSEGF
TRLWPSVGTIICYCASFWLLAQTLAYIP
TGIAYAIWSGVGIVLISLLSWGFFGQRL
DLPAIIGMMLICAGVLIINLLSRSTPH



MSEALKILNNIRTLRAQARECTLETLEEMLE
KLEVVVNERREEESAAAAEVEERTRKLQY
REMLIADGIDPNELLNSLA AVKSGTKAKRA
QRPAYSYVDENGETKTWTGQGRTPAVIK



To align two sequences of 300 aa/nt lengths,
approximately 10^{88} comparisons are to be made

EMBOSS Programs

Feedback

Share

Tools > EMBOSS Programs

Selected EMBOSS tools for sequence analysis

Pairwise Sequence Alignment

Needle ?

Create an optimal global alignment of two sequences using the Needleman-Wunsch algorithm

Protein Nucleotide

Stretcher ?

Improved version of the Needleman-Wunsch algorithm that allows larger sequences to be globally aligned

Protein Nucleotide

Water ?

Use the Smith-Waterman algorithm to calculate the local alignment of two sequences

Protein Nucleotide

Matcher ?

Emboss homepage

Protein alignment
Nucleotide alignment
Web services
Help & Documentation
Feedback
Share

This is the form for protein sequences. Please go to the [nucleotide form](#) if you wish to align DNA or RNA sequences.

STEP 1 - Enter your protein sequences

Enter or paste your first **protein** sequence in any supported format:

```
>Flavohemoprotein_Escherichia_coli
MLDAQTIATVKATIPLLVETGPKLTAHFYDRMFTHNPELKEIFNMSNQNRNGDQREALFNA
IAAYASNIENTPALLPAVEKIAQKHTSFQIKPEQYNIVGEHLLATLDEMSPGQEVLDW
GKAYGVLNVFINREAEIYNENASKAGGWEGTRDFRIVAKTPRSALITSFELEPVDGGAV
AEYRPGQYLGVWLKPEGFPHQEIRQYSLTRKPDGKGYRIAVKREEGGQVSNWLNHANVG
DVVKLVAPAGDFFMAVADDTPTVLISAGVGQTPMLAMLDLAKAGHTAQVNWFFHAAENG
VHAFADDEVKELGQSLPRFTAHTWYRQPSEADRAKGQFDSEGLMDLSKLEGAFSPTMQFY
LCGPVGFQMFTAQQLVDLGVKQENIHYESCFGPHKVL
```

Or, [upload a file](#): No file chosen

AND

Enter or paste your second **protein** sequence in any supported format:

```
>Hemoglobin_subunit_alpha_Homo_sapiens
MVLSPADKTNVKAAGWGVGAHAGEYGAEALERMFLSFPTTKTYFPHFDLSHGSAQVKGHG
KKVADALTNAVAHVDDMPNALSALSDLHAHKLRLVDPVNFKLLSHCLLVTLAAHLPAEFTP
AVHASLDKFLASVSTVLTSKYR
```

Or, [upload a file](#): No file chosen

STEP 2 - Set your pairwise alignment options

Entering the sequence to be compared in
N-W algorithm

Protein alignment	Nucleotide alignment	Web services	Help & Documentation	Feedback	Share
#	#	#	#	#	#
Flavohemoprot	1 M-LDAQTIATVKATIPLLVETGPKLTAHFYDRMFTHNPKEIFNMSNQR	49			
Hemoglobin_su	1 MVLSPADKTIIVKAAWGVGAHAGEYGAEALERMFSPPTTKTYFPFDLS	50			
Flavohemoprot	50 NGDQR-----EALFNAIAAYASNIENLPALLPAVEKI-AQKHTSFQ	89			
Hemoglobin_su	51 HGSAAQVKGHGKGVADALTNVA---HVDDMPNALSALSDLHAHK---LR	93			
Flavohemoprot	90 IKPEQYNIIVGEHLATLDEMFSQGEVLDANGKAYGVLANVFINREAEIY	139			
Hemoglobin_su	94 VDPVNFKLLSHCLLVTL-----	110			
Flavohemoprot	140 NENASKAGGWEGTRDFRIVAKTPRSALITSFELEPVDGGAVAEYRPGQYL	189			
Hemoglobin_su	111 -----AAHLP-----AEFTPA---	121			
Flavohemoprot	190 GVNILKPEGFPHQEIQRQYSLTRKPDGKGYRIAVKREEGQVSNMLHNIHANV	239			
Hemoglobin_su	122 -----	121			
Flavohemoprot	240 GDVVKLVPAGDFFMAVADDTPTVLISAGVGQTPMLAMLDLAKAGHTAQ	289			
Hemoglobin_su	122 -----	121			
Flavohemoprot	290 VMNFHAAENGVDVHFADEVKELGQSLPRFTAHTWYRQPSADRAKGQFDS	339			
Hemoglobin_su	122 -----VHA-----SLDKFLA-----S	132			
Flavohemoprot	340 EGLMDLSKLEGAFSDPTMQFYLCGPVGMQFTAKQLVDLGVKQENIHVEC	389			
Hemoglobin_su	133 VSTVLTSKYR-----	142			
Flavohemoprot	390 FGPHKVL	396			
Hemoglobin_su	143 -----	142			
#	#	#	#	#	#

Protein alignment result

Pairwise sequence alignment

- When comparing 2 sequences it is **Pairwise sequence alignment** (nucleic acids or protein)
- When comparing more than 2 sequence it is **Multiple sequence alignment** (nucleic acids or protein)

Contd...

- Pairwise sequence alignment is concerned with comparing 2 DNA or 2 Amino acids sequences
- For ex.
 - BLASTn :- for nucleotide sequence
 - BLASTp :- for protein sequence

BLAST-n and BLAST-P

- BLAST is a **B**asic **L**ocal **A**lignment **S**earch **T**ool.
- Used for comparing primary biological information viz,
 - Amino acid sequences of protein
 - Nucleotide of DNA or RNA sequences
- BLASTn and BLASTp is particularly used for comparing nucleotide sequence and protein sequence respectively

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BLAST Results

[Edit and Resubmit](#) [Save Search Strategies](#) [Formatting options](#) [Download](#) [YouTube](#) [How to read this page](#) [Blast report description](#)

Job title: ref|NP_001276427| (867 letters)

RID [VNKBX2DD015](#) (Expires on 10-09 11:22 am)

Query ID [NP_001276427.1](#)
 Description pikachurin isoform 3 [Mus musculus]
 Molecule type amino acid
 Query Length 867

Database Name nr
 Description All non-redundant GenBank CDS translations+PDB+SwissProt+PIR+PRF excluding environmental samples from WGS projects
 Program BLASTP 2.8.1+ [Citation](#)

Other reports: [Search Summary](#) [Taxonomy reports](#) [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#)

New Analyze your query with [SmartBLAST](#)

Graphic Summary

[Show Conserved Domains](#)

Putative conserved domains have been detected, click on the image below for detailed results.

Query seq. 1 125 250 375 500 625 750 867

Cytokine receptor motif

Specific hits FN3 FN3 super LanG LanG superfamily EGF_CA EGF superfamily EGF_C Laminin_6_2

Superfamilies FN3 super LanG superfamily EGF_CA EGF superfamily EGF_C LanG superfamily

Distribution of the top 100 Blast Hits on 100 subject sequences ⓘ

Mouse over to see the title, click to show alignments

Color key for alignment scores

<40 40-50 50-80 80-200 >=200

[Questions/comments](#)

Result of *Paralichthys olivaceus* in BLASTn

Sequences producing significant alignments:

Select: [All](#) [None](#) Selected: 0

[Alignments](#) [Download](#) [GenBank](#) [Graphics](#) [Distance tree of results](#)

	Description	Max score	Total score	Query cover	E value	Ident	Accession
☐	Paralichthys olivaceus mRNA for keratin, partial cds	662	662	100%	0.0	100%	AB049616.1
☐	PREDICTED: Paralichthys olivaceus keratin, type I cytoskeletal 13-like (LOC109627494), mRNA	651	651	100%	0.0	99%	XM_020084024.1
☐	PREDICTED: Lates calcarifer keratin, type I cytoskeletal 13-like (LOC108875644), mRNA	401	401	100%	7e-108	87%	XM_018664684.1
☐	PREDICTED: Seriola dumerili keratin, type I cytoskeletal 13-like (LOC111229213), mRNA	385	385	100%	7e-103	86%	XM_022755440.1
☐	PREDICTED: Mastacembelus armatus keratin, type I cytoskeletal 50 kDa-like (LOC113128586), mRNA	318	318	100%	7e-83	83%	XM_026304028.1
☐	PREDICTED: Mastacembelus armatus keratin, type I cytoskeletal 13-like (LOC113128588), mRNA	307	307	100%	2e-79	82%	XM_026304029.1
☐	Gasterosteus aculeatus clone CFW18-C12 mRNA sequence	296	296	100%	3e-76	82%	BT026740.1
☐	PREDICTED: Stegastes partitus keratin, type I cytoskeletal 13-like (LOC103362534), mRNA	285	285	100%	7e-73	81%	XM_008288915.1
☐	PREDICTED: Amphiprion ocellaris keratin, type I cytoskeletal 13-like (LOC111566298), mRNA	274	274	100%	2e-69	81%	XM_023266822.1
☐	PREDICTED: Austrofundulus limnaeus keratin, type I cytoskeletal 13-like (LOC106518778), mRNA	202	202	100%	8e-48	77%	XM_014010178.1
☐	PREDICTED: Kryptolebias marmoratus keratin, type I cytoskeletal 13-like (LOC108236409), mRNA	185	185	100%	8e-43	76%	XM_017417037.2
☐	PREDICTED: Hipposideros armiger keratin, type I cytoskeletal 14 (LOC109395737), mRNA	141	141	68%	2e-29	78%	XM_019667004.1
☐	PREDICTED: Octodon degus keratin, type I cytoskeletal 14 (LOC101569745), mRNA	135	135	68%	8e-28	77%	XM_004633755.2
☐	PREDICTED: Microcebus murinus keratin, type I cytoskeletal 14 (LOC105872035), mRNA	135	135	68%	8e-28	78%	XM_012765776.2
☐	PREDICTED: Acinonyx jubatus keratin, type I cytoskeletal 17-like (LOC106972591), mRNA	135	135	68%	8e-28	77%	XM_015069589.1
☐	PREDICTED: Jaculus jaculus keratin, type I cytoskeletal 14 (LOC101596804), transcript variant X2, mRNA	135	135	68%	8e-28	77%	XM_004655400.1
☐	PREDICTED: Jaculus jaculus keratin, type I cytoskeletal 14 (LOC101596804), transcript variant X1, mRNA	135	135	68%	8e-28	77%	XM_004655399.1
☐	PREDICTED: Ceratotherium simum simum keratin, type I cytoskeletal 14 (LOC101406817), mRNA	135	135	68%	8e-28	77%	XM_004434491.1
☐	PREDICTED: Mus pahari keratin, type I cytoskeletal 14 (LOC110331109), transcript variant X2, mRNA	130	130	68%	4e-26	77%	XM_021211493.1
☐	PREDICTED: Mus pahari keratin, type I cytoskeletal 14 (LOC110331109), transcript variant X1, mRNA	130	130	68%	4e-26		

[Questions/comm](#)

Sequence producing significant alignment

Download ▾
[GenBank](#)
[Graphics](#)

Paralichthys olivaceus mRNA for keratin, partial cds
Sequence ID: [AB049616.1](#) Length: 358 Number of Matches: 1

Range 1: 1 to 358 [GenBank](#) [Graphics](#)

	Score	Expect	Identities	Gaps	Strand
	662 bits(358)	0.0	358/358(100%)	0/358(0%)	Plus/Plus

Query	1	AAGAACCAGAGAGAACTTGAAGCCTGGCTCCAGACACAGTCAGAGTCGCTGAGCAAGGAG	60
Sbjct	1	AAGAACCAGAGAGAACTTGAAGCCTGGCTCCAGACACAGTCAGAGTCGCTGAGCAAGGAG	60
Query	61	GTGGCAGTCAAGACAGAAAGTTCTCAAACGACCAAGGCAGAAATCTCTGACCTCCGTCGC	120
Sbjct	61	GTGGCAGTCAAGACAGAAAGTTCTCAAACGACCAAGGCAGAAATCTCTGACCTCCGTCGC	120
Query	121	ACAATGCAGAACCTGGAGATCGAGCTGCAGTCTCACTTAGCATGAAAGCGGCACTGGAA	180
Sbjct	121	ACAATGCAGAACCTGGAGATCGAGCTGCAGTCTCACTTAGCATGAAAGCGGCACTGGAA	180
Query	181	GGAACGTTGTCAGACACAGAGTGTCTTACTCCATGCAGCTCAAAGCCTCCAGATACAG	240
Sbjct	181	GGAACGTTGTCAGACACAGAGTGTCTTACTCCATGCAGCTCAAAGCCTCCAGATACAG	240
Query	241	GTGACGAGCCTGGAAGAACAGCTGATGCAGCTGCGTCCGACATGGAGCGCCAGAACCAA	300
Sbjct	241	GTGACGAGCCTGGAAGAACAGCTGATGCAGCTGCGTCCGACATGGAGCGCCAGAACCAA	300
Query	301	GAGTACAACATCCTGCTCGACATCAAGACACGGCTGGAGATGGAAATTGCAGAATACA	358
Sbjct	301	GAGTACAACATCCTGCTCGACATCAAGACACGGCTGGAGATGGAAATTGCAGAATACA	358

Download ▾
[GenBank](#)
[Graphics](#)

PREDICTED: Paralichthys olivaceus keratin, type I cytoskeletal 13-like (LOC109627494), mRNA
Sequence ID: [XM_020084024.1](#) Length: 1471 Number of Matches: 1

Range 1: 859 to 1216 [GenBank](#) [Graphics](#)

	Score	Expect	Identities	Gaps	Strand
	651 bits(352)	0.0	356/358(99%)	0/358(0%)	Plus/Plus

Query	1	AAGAACCAGAGAGAACTTGAAGCCTGGCTCCAGACACAGTCAGAGTCGCTGAGCAAGGAG	60
Sbjct	859	AAGAACCAGAGAGAACTTGAAGCCTGGTTCCAGACACAGTCAGAGTCGCTGAGCAAGGAG	918
Query	61	GTGGCAGTCAAGACAGAAAGTTCTCAAACGACCAAGGCAGAAATCTCTGACCTCCGTCGC	120

[Gene](#) - associated gene details
[GEO Profiles](#) - microarray expression data

[Gene](#) - associated gene details
[New Genome Data Viewer](#) - aligned genomic context

Questions/commen

Sequence alignment of *Paralichthys olivaceus*

Applications of pairwise sequence alignment

- ✓ Searching large sequences for matches
- ✓ Characterize newly sequenced genes or gene products
- ✓ Molecular distance of evolution between species

Lunch Break

Day 2: Sequence Alignment and Phylogenetic

Afternoon Session

Outline Day 2

Afternoon (2-5 pm): Phylogenetics

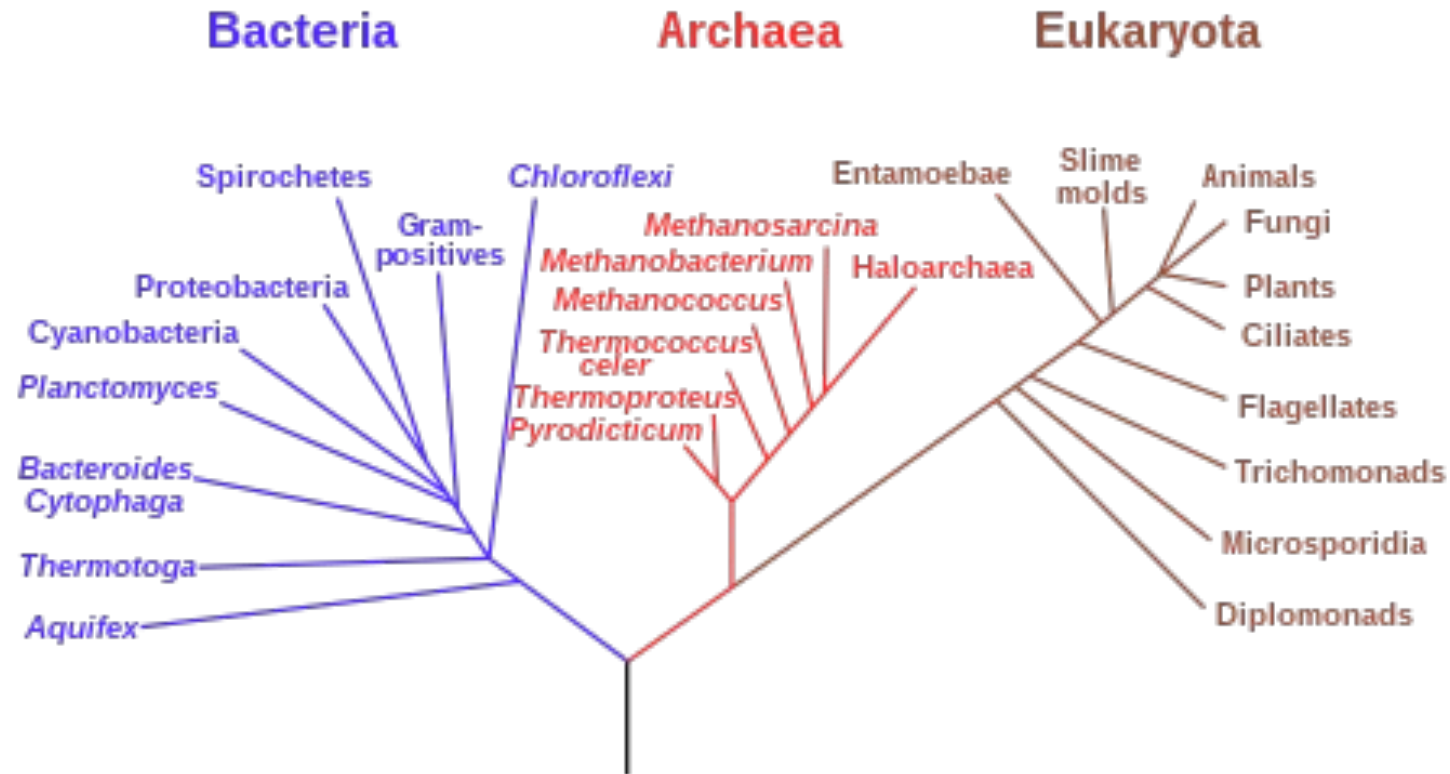
Phylogenetics:

- Introduction to Phylogenetics tree concept, and significance in evolution biology.

Practical Session:

- Construct phylogenetics trees using Clustal Omega (web-based) starting from multiple sequence alignments.
- Code with python

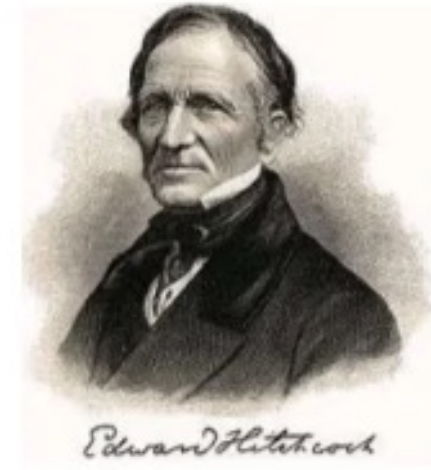
Phylogenetic tree



- A **phylogenetic tree** is a diagram that represents evolutionary relationships among organisms based on the similarities and differences in their genetic and evolutionary characteristics
- The pattern of branching in a phylogenetic tree reflects how species or other groups evolved from a series of common ancestors.
- The phylogenetic tree is also called the “Tree of Life” or “Dendrogram”

History

- Early representations of "branching" phylogenetic trees include a "paleontological chart" showing the geological relationships among plants and animals in the book *Elementary Geology*, by Edward Hitchcock in 1840.
- Charles Darwin in 1859 also produced one of the first illustrations and crucially popularized the notion of an evolutionary "tree" in his seminal book *The Origin of Species*.



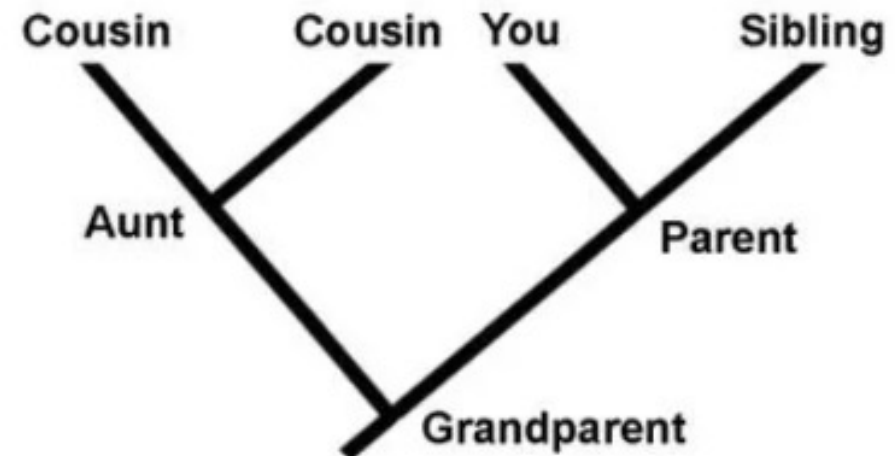
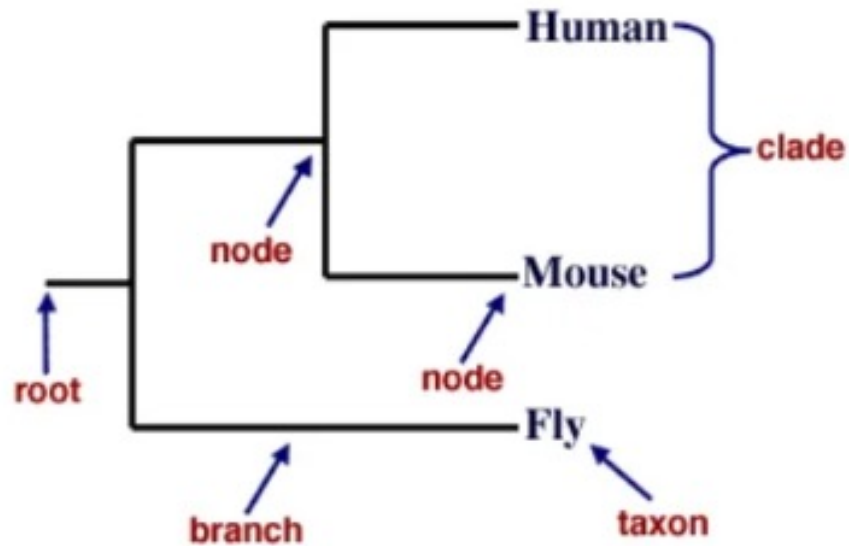
Importance of Phylogenetic Tree

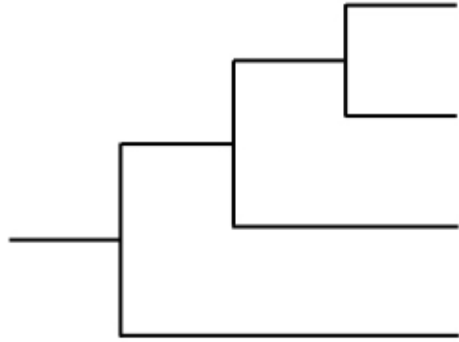
- It is the fundamental tool to derive their most-useful evidence from the fields of anatomy, embryology, palaeontology and molecular genetics. Other significances of the phylogenetic tree are:
- Used in the search for a new species.
- Used to study evolutionary histories.
- To study how the species were spread geographically.
- To study the common ancestors of extant and extinct species.
- It is used to identify the most recent common ancestors and to recognize how closely related species are.
- To relate the milestones of the evolution of major life forms to the tree of life.
- To represent evolutionary relationships between organisms that are believed to have some common ancestry.
- With the help of the phylogenetic tree, the infectious microbes can be traced along with their evolutionary histories.

Term	Explanation
Phylogeny	A method to construct a phylogenetic tree
Common ancestors	A group of organisms sharing the common feature with the decedent.
Taxon	An organism of the entire taxa
Taxa	A group of organisms from a species or from a different species.
Node	Nodes represent the common ancestors of Different taxons.
Sister group	Two or more taxon share the same node.
Outgroup	The taxon is outside the interest group or other than the common ancestor.
Clade	A clade is a group of all organisms from a common ancestor.

What does this tree looks like?

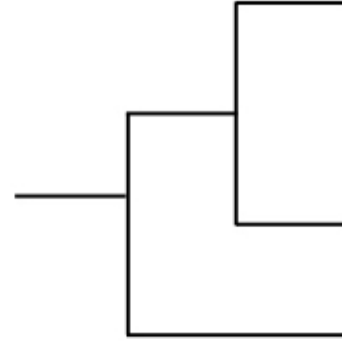
➤ What do the lines represent?





A

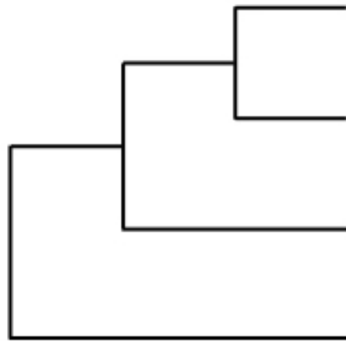
Rooted tree



C

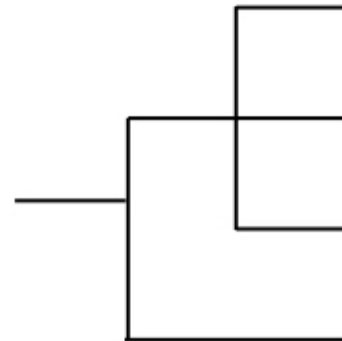
Bifurcating tree

© Genetic Education Inc.



B

Unrooted tree

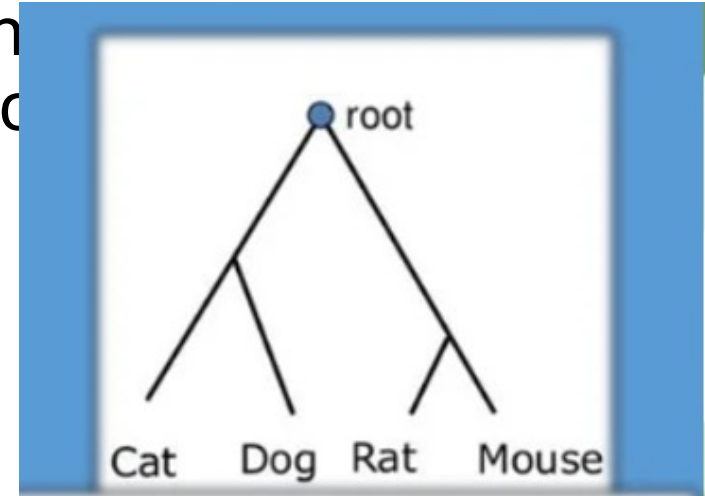


D

Multifurcating tree

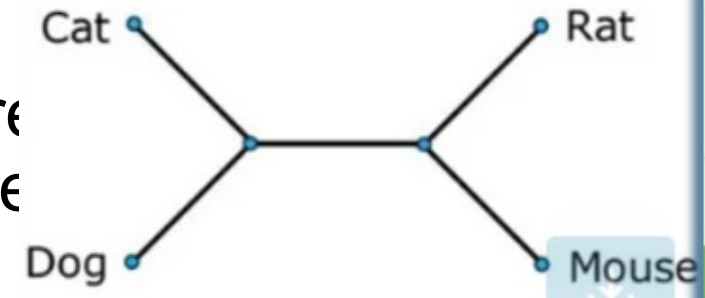
Rooted tree:

- The rooted tree is described as a phylogenetic tree sharing the common ancestor on the node. Therefore the classification ends at one point usually on the node which is the common ancestor of all the branches of the tree.



Unrooted tree:

- Contrary to the rooted tree, the non-rooted tree doesn't have a common ancestor. The unrooted phylogenetic tree is always prepared from the rooted tree by excluding the common ancestor or the node of the tree.



Bifurcating tree

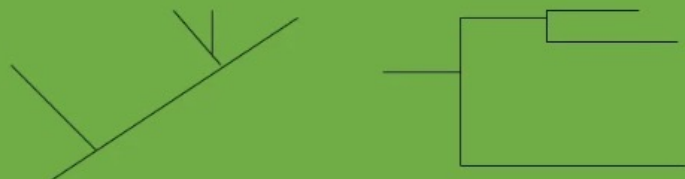
- The phylogenetic tree only has two branches or we can say leaves are known as bifurcating trees. It is also classified in rooted bifurcation trees and unrooted bifurcating trees.

Multifurcating tree:

- The multifurcating tree is described as having multiple branches on a single node. Again, it is classified into a rooted multifurcating tree and an unrooted multifurcating tree.

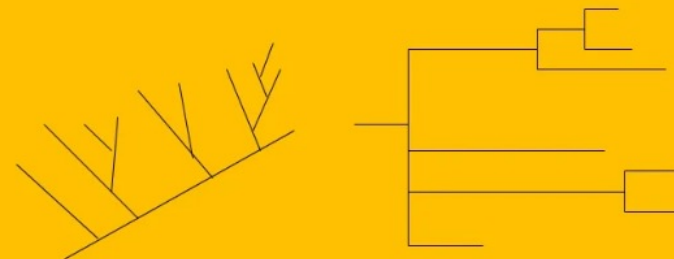
The Bifurcating Tree

- A tree that bifurcates has a maximum of 2 descendants arising from each of the interior nodes.



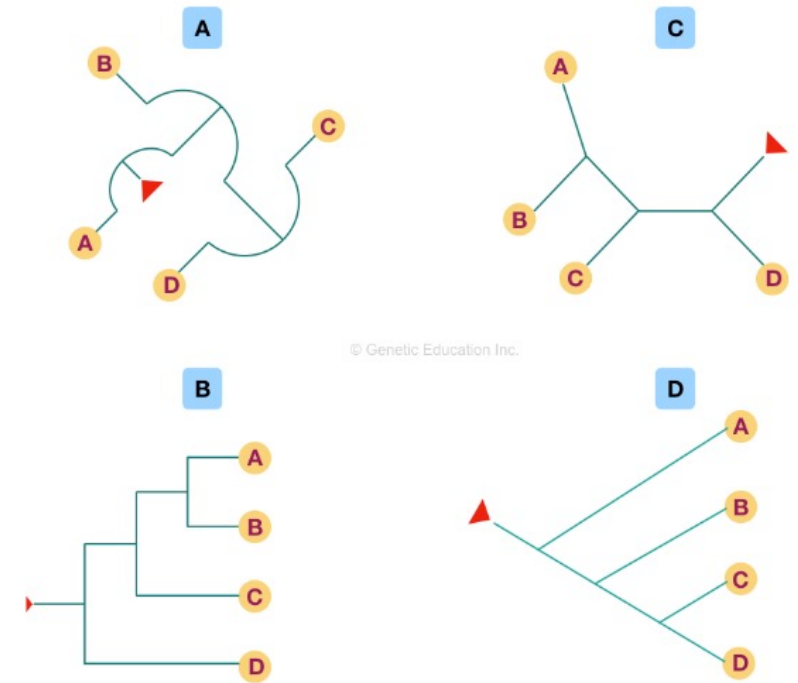
The Multi-furcating Tree

- A tree that multi-furcates has multiple descendants arising from each of the interior nodes.



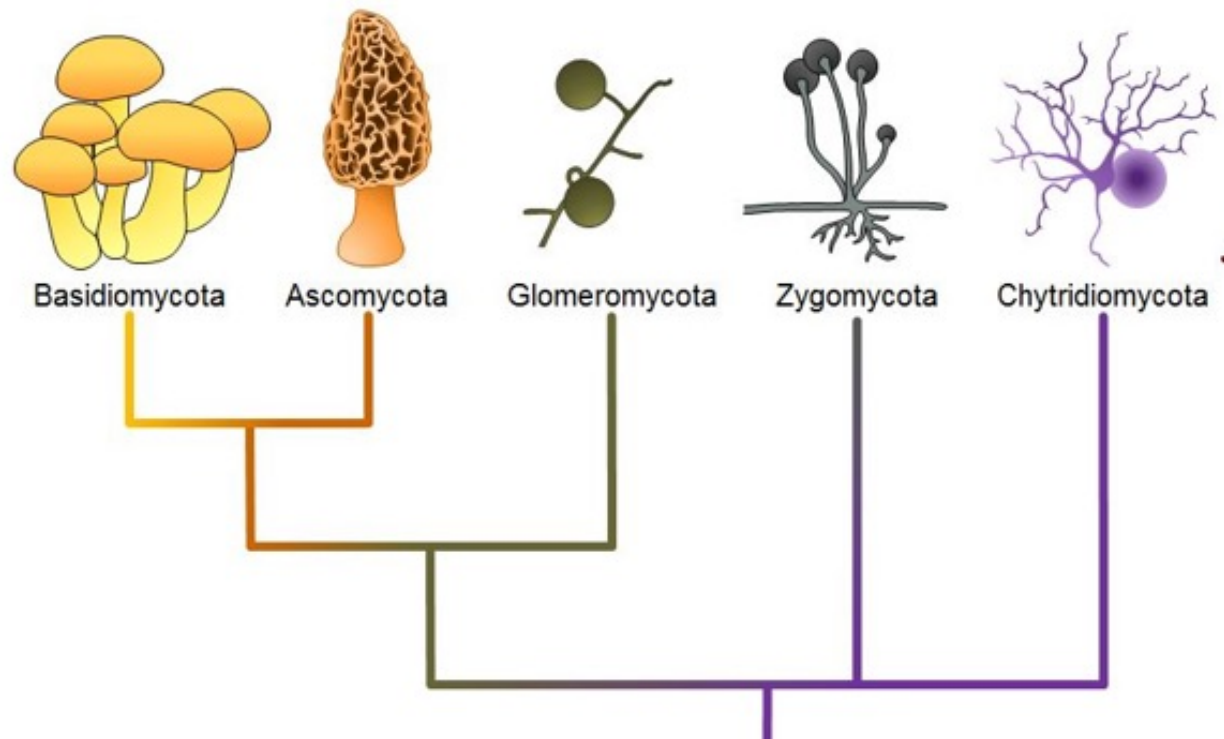
In phylogeny, the node is also known as a “clade” as well. Though there are so many different variations of the phylogenetic tree, every method of making a tree depicts the same type of information. Take a look at various trees shown in the figure

Keep in mind that whatever the shape, topology or structure of the tree is, it must have a common node if rooted and branched. To read a tree, start with the tip of the branches and see where the branch ends (the node), based on that information you can depict or conclude which organism is nearer or closer and which are distantly related.



The figure represents different forms of a single type of phylogenetic tree.

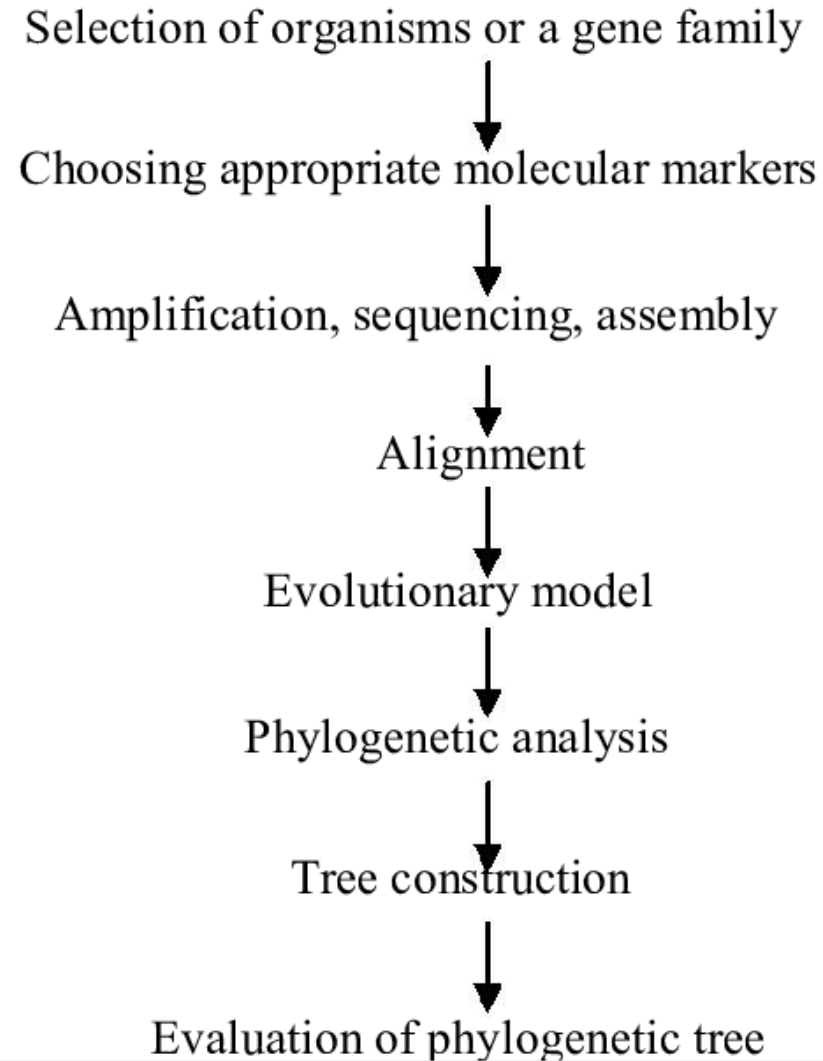
Phylogeny in fungus



Applications of a phylogenetic tree:

- The phylogenetic tree is constructed to make an evolutionary link between various organisms. By doing so, we can get an idea about how and from whom different organisms are evolved.
- Also, it helps to classify organisms and species in different taxa and groups based on their DNA sequence and phenotypic similarities and differences.
- In addition to this, it is useful to study the force of evolution and characteristics of different organisms.
- it is applicable to study the events occurring during the course of evolution and to classify species based on the divergence of structure and function.

Steps in Phylogenetic Analysis



Software

- Some programs for phylogenetic analysis
- A multiple alignment program:
 - Clustal, T-Coffee, MAFFT, Muscle...
- A phylogenetic program:
 - Phylip, PAUP*, MacClade, BioEdit...
- Visualizing the tree:
 - TreeView, Njplot
- **<https://evolution.genetics.washington.edu/phylip/software.html>**

Selecting sequence

- The rate of mutation is assumed to be the same in both coding and non-coding region
- However there is a difference in substitution rate
- Non-coding DNA region have more substitution than coding regions.
- Protein are much more conserved since they need to conserve their function
- It is better to use sequence that mutate slowly (protein) than DNA. If the gene are very small or they mutate slowly, then it can be used for building tree.

Building Phylogenetic Trees

- The most popular and frequently used methods of tree building can be classified into two major categories
- Phenetic methods based on **distances**
- Cladistic methods based on **characters**
- **Distance matrix methods**
 - UPGMA (Unweighted pair group methods with arithmetic mean)
 - Fitech-Margoliash
 - Neighbour joining (NJ)
- **Character based methods**
 - Maximum parsimony (MP)
 - Maximum likelihood (ML)

Distance based methods

- Tree are calculated by similarities of sequences and are based on distance
- Some sequences more similar than others
- Closely related sequences should be close in the tree
- Only use the distances between sequences
- All methods start with a *distance matrix*

UPGMA Vs Fitch Margoliash Method

UPGMA Methods

Unweighted Pair Group Method with Arithmetic Mean

Unweighted: The distances are used as they are

Pair: Find the two closest elements

Group: Put them together in a new group

Arithmetic Mean: Gives distances from the new group

Fitch-Margoliash Methods

More complicated than UPGMA

Does not assume a molecular clock

Produces an unrooted scaled tree

Neighbour joining

- This method tries to correct the UPGMA method for its assumption that the rate of evolution is the same in all taxa.
- But it assumes an additive tree
 - Distance between two leaves is the sum of the edges
- Find the *closest pair* that is *most apart* from the rest of the tree
- Connect pair and update distances
 - A little advanced: Take the overall distance to the rest of the tree into account
 - Corrects for varying mutation
- Fast and can give good results

Character methods

- Tree are calculated by considering the various possible pathways of evolution.
- This methods uses each alignment positions as evolutionary information to build a tree.
- All information at hand
- More advanced, slower, but also more accurate
- Maximum Parsimony (MP)
 - Occam's razor: Simplest explanation
- Maximum Likelihood (ML)
 - Advanced statistical method
 - Most probable tree given the data and the model

Maximum parsimony (MP)

- For each position in the alignment all possible trees are evaluated and are given a score based on the number of evolutionary changes.
- More time consuming
- Used for closely related sequences
- The most parsimonious tree is the one with the fewest evolutionary changes
- MP methods are available for DNA in Programs **paup**, **molphy**, **phylo_win**

Maximum Likelihood

- This method also uses each position in an alignment, evaluate all possible trees and calculates the likelihood for each tree.
- The tree with the maximum likelihood is the most probable tree.
- Slowest method but gives the best result
- Used for any set of sequence.
- Maximum likelihood methods can be found in **phylip, paup or puzzle**